Structured Representation Learning for Low-Resource Speech Coding

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1 Problem Statement

We investigate whether audio admits a compact, structured, and reconstructible representation whose constituent information can be separated, organized, related, and efficiently encoded.

Unlike conventional neural codecs, which primarily optimize a latent representation for rate–distortion performance, our objective is to investigate the structure of the representation itself. Specifically, we seek to determine what constituent information is required to describe speech, how this information can be organized into distinct representations, and how these representations can jointly reconstruct the original signal.

We propose representing speech through a finite collection of *feature slots*. Each slot acts as a container for a particular class of information extracted from the input. The contents and relationships of these slots collectively constitute the representation of the audio.

Importantly, the constituent features need not be statistically independent or mutually orthogonal. Instead, the representation should minimize unnecessary redundancy while retaining sufficient information for reconstruction.

The long-term objective is to discover these constituent features automatically. However, our initial investigation uses known acoustic properties of human speech to establish the feasibility of the proposed representation.

2 Feature-Based Speech Representation

Consider an audio signal divided into T frames,

$$X = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}, \qquad \mathbf{x}_t \in \mathbb{R}^N. \tag{1}$$

We introduce K feature slots,

$$\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_K\}, \tag{2}$$

where each slot contains information extracted from the audio through a corresponding analysis operator.

For the initial supervised formulation, we define

$$\boxed{\mathbf{z}_{k,t} = E_k(\mathbf{x}_{\leq t}; \theta_k)} \tag{3}$$

where E_k is a learned feature extractor, θ_k denotes its parameters, and $\mathbf{z}_{k,t} \in \mathcal{Z}_k$ is the representation associated with slot k at time t.

Each slot therefore contains a time-indexed representation,

$$\mathcal{C}_k(X) = \{\mathbf{z}_{k,1}, \ldots, \mathbf{z}_{k,T}\}. \tag{4}$$

The complete structured representation is

$$\boxed{\mathcal{Z}(X) = \{\mathcal{C}_1(X), \ldots, \mathcal{C}_K(X)\}.} \tag{5}$$

The individual latent spaces $\mathcal{Z}_k$ need not have the same dimensionality or represent the same type of acoustic information.

2.1 Initial Feature-Slot Definitions

For the first experiment, we consider the following known acoustic properties of human speech:

1. **Pitch and voicing:** Fundamental-frequency contours and the presence or absence of periodic excitation.
2. **Spectral envelope:** Slowly varying spectral characteristics associated with vocal-tract filtering and formant structure.
3. **Aperiodicity:** Noise-like excitation components and departures from harmonic periodicity.
4. **Transient structure:** Rapidly varying acoustic events associated with articulation, consonant bursts, and temporal changes.

These are not assumed to be independent components. For example, pitch and voicing are closely related, while spectral-envelope characteristics contribute to both phonetic identity and perceived timbre.

An additional residual representation may be introduced to capture information inadequately represented by the initial feature slots.

The purpose of this initial selection is not to claim that these properties constitute an optimal decomposition of speech, but to establish a measurable starting point.

3 Feature Extraction and Reconstruction

Each feature extractor is initially trained to estimate its corresponding acoustic parameters.

Given reference feature representations $\mathbf{z}^*_{k,t}$ obtained from established signal-processing methods, a preliminary training objective is

$$\mathcal{L}_{\text{feat}} = \sum_{k=1}^{K} \alpha_k \mathbb{E}_t \left[d_k \left(E_k(\mathbf{x}_{\leq t}), \mathbf{z}^*_{k,t} \right) \right], \tag{6}$$

where d_k is an appropriate feature-specific distortion measure.

The extracted representations are combined using a shared synthesis operator S:

$$\boxed{\hat{\mathbf{x}}_t = S(\mathbf{z}_{1,\leq t}, \ldots, \mathbf{z}_{K,\leq t}).} \tag{7}$$

The first objective is to establish whether the selected representations contain sufficient information for reconstructing speech before introducing quantization.

This can initially be evaluated using classical analysis–synthesis systems, such as WORLD [2], which represents speech through fundamental frequency, spectral envelope, and aperiodicity.

Once individual extraction performance has been established, the extractors and synthesizer may be jointly fine-tuned using a reconstruction objective,

$$\mathcal{L}_{\text{rec}} = \mathbb{E}\left[D\left(\mathbf{x}, \hat{\mathbf{x}} \right) \right], \tag{8}$$

where D may combine waveform and perceptually motivated spectral distortion measures.

This distinction is important: accurate estimation of individual acoustic parameters does not necessarily guarantee accurate joint waveform reconstruction.

4 Quantization and Transmission

Having established a reconstructible representation, we investigate how efficiently its constituent information can be transmitted.

Each feature representation is quantized independently in the initial experiment:

$$\tilde{\mathbf{z}}_{k,t} = Q_k(\mathbf{z}_{k,t}). \tag{9}$$

The decoder reconstructs the audio using only the received quantized representations:

$$\boxed{\hat{\mathbf{x}}_t = S(\tilde{\mathbf{z}}_{1,\leq t}, \ldots, \tilde{\mathbf{z}}_{K,\leq t}).} \tag{10}$$

Because the initial slots have fixed, predefined meanings, their identities need not be transmitted explicitly. The bitstream instead contains the quantized feature parameters and any necessary timing, scaling, or activity information.

For a fixed-rate codec, the total transmitted bits over a coding interval must satisfy

$$\boxed{B_{\text{total}} = \sum_{k=1}^{K} B_k + B_{\text{overhead}} \leq B_{\max}.} \tag{11}$$

We therefore investigate the constrained rate–distortion problem

$$\boxed{\begin{aligned} &\min_{\{Q_k\},S} \quad \mathbb{E}[D(\mathbf{x}, \hat{\mathbf{x}})] \\ &\text{subject to} \quad B_{\text{total}} \leq B_{\max}. \end{aligned}} \tag{12}$$

Different slots may require different quantization strategies and bit allocations. The objective is to establish the marginal reconstruction benefit of allocating bits to each feature family.

If the initial feature representation is insufficient, an additional learned residual slot may be introduced to encode information missing from the base reconstruction.

5 Extension to Learned Slot Discovery

The preceding formulation assumes that the contents of each slot are known beforehand.

Our longer-term objective is to remove this assumption and allow the architecture to discover useful constituent representations automatically.

We propose first extracting a collection of learned feature tokens from each audio frame,

$$\mathcal{H}_t = \{\mathbf{h}_{t,1}, \ldots, \mathbf{h}_{t,L}\}. \tag{13}$$

A learned assignment mechanism then determines the association between each token and the available slots:

$$\boxed{a_{t,n,k} = P(\mathcal{C}_k \mid \mathbf{h}_{t,n}), \qquad \sum_{k=1}^{K} a_{t,n,k} = 1.} \tag{14}$$

The representation of slot k is obtained by aggregating its assigned information,

$$\mathbf{z}_{k,t} = \text{Agg}_k \left(\{a_{t,n,k} \mathbf{h}_{t,n}\}_{n=1}^{L} \right). \tag{15}$$

This formulation is related to slot-based representation learning, including Slot Attention [1], but differs in its intended application: rather than identifying discrete objects or sound sources, we seek constituent acoustic information suitable for reconstruction and efficient coding.

Learned slot assignments may require additional transmission information. Their coding cost must therefore be included when evaluating the representation.

6 Functional Separation and Statistical Dependencies

A major challenge is preventing different slots from learning redundant representations.

However, conventional orthogonality between dictionary matrices,

$$\mathbf{D}_i^{\top} \mathbf{D}_j = \mathbf{0}, \tag{16}$$

may be unnecessarily restrictive because different speech properties are neither physically nor statistically independent.

Simon et al. [3] propose an alternative principle based on the functional effects of latent variables on a generative mapping.

For a differentiable synthesis function $\hat{\mathbf{x}} = S(\mathbf{z})$, define

$$\mathbf{J}_{S,k} = \frac{\partial S}{\partial \mathbf{z}_k}. \tag{17}$$

A possible group-wise functional orthogonality regularizer is

$$\boxed{\mathcal{L}_{\text{func}} = \sum_{i<j} \left\| \mathbf{J}_{S,i}^{\top} \mathbf{J}_{S,j} \right\|_F^2 .} \tag{18}$$

This encourages changes in different latent groups to produce complementary local changes in the reconstructed waveform.

It is an adaptation of the functional-orthogonality principle, not a direct application of Simon et al.'s identifiability theorem, which requires additional assumptions on the generative mapping and latent domain.

Functional separation also does not imply statistical independence. Consequently, dependencies between slot representations may still be exploited for coding.

For example, a predictive model could estimate

$$\hat{\mathbf{z}}_{j,t} = g_j \left(\{ \tilde{\mathbf{z}}_{i,\leq t} : i \in \text{Pa}(j) \} \right), \tag{19}$$

where the parent representations are already available to the decoder.

Only the prediction innovation would then be encoded:

$$\boxed{\begin{aligned} \mathbf{e}_{j,t} &= \mathbf{z}_{j,t} - \hat{\mathbf{z}}_{j,t}, \\ \tilde{\mathbf{z}}_{j,t} &= \hat{\mathbf{z}}_{j,t} + Q_j(\mathbf{e}_{j,t}). \end{aligned}} \tag{20}$$

Such dependencies may eventually be represented using a sparse directed graph. Graph learning and functional orthogonality are therefore considered subsequent extensions rather than prerequisites for the initial representation experiment.

7 Proposed Research Methodology

The investigation is organized into four stages.

1. **Known-property representation:** Extract pitch, voicing, spectral-envelope, and aperiodicity information using established analytical methods. Evaluate reconstruction before quantization.

2. **Learned feature extraction and coding:** Train lightweight feature-specific networks and a shared synthesizer. Quantize the representations and evaluate reconstruction at matched bitrates and computational costs.

3. **Unsupervised slot discovery:** Replace predefined feature assignments with a learned routing mechanism. Investigate representation sufficiency, slot utilization, stability, and redundancy.

4. **Functional and statistical structure:** Evaluate functional orthogonality constraints and subsequently investigate predictive relationships between slots for reducing transmitted information.

Throughout these experiments, three properties remain central: reconstruction fidelity, representation economy, and computational efficiency.

The ultimate research question is whether a structured representation of audio, built from learned constituent information and its relationships, can provide a better rate–distortion–complexity tradeoff than an unconstrained latent representation.

The objective is not to assume that disentanglement or orthogonality improves compression, but to establish whether organizing audio information into such representations provides a measurable advantage.

References

[1] F. Locatello et al., "Object-Centric Learning with Slot Attention," *Advances in Neural Information Processing Systems*, 2020.

[2] M. Morise, F. Yokomori, and K. Ozawa, "WORLD: A Vocoder-Based High-Quality Speech Synthesis System for Real-Time Applications," *IEICE Transactions on Information and Systems*, 2016.

[3] M. C. Simon, P. Frossard, and C. De Vleeschouwer, "Unsupervised Disentanglement Without Compromises: How Functional Orthogonality Enforces Identifiability," *Proceedings of the 43rd International Conference on Machine Learning*, 2026.